CENG381 Stochastic Processes

Biased Random Walk - Answer Key 1

Q1 Solution

a) Recurrence and boundary conditions:

Conditioning on the first step from position k:

$R(k) = p*R(k+1) + q*R(k-1)$ for $k = 1, 2, 3, 4$

$R(0) = 0$ (gambler is ruined; winning probability = 0)

$R(5) = 1$ (gambler has won all 5 units; winning prob = 1)

b) Characteristic equation and general solution:

Substitute $R(k) = r^k$ into $p*R(k+1) - R(k) + q*R(k-1) = 0$:

$p*r^{(k+1)} - r^k + q*r^{(k-1)} = 0$

Divide by $r^{(k-1)}$: $p*r^2 - r + q = 0$

Using the quadratic formula with p=0.4, q=0.6:

$r = (1 +/- \text{sqrt}(1 - 4pq)) / (2p)$

$4pq = 4*0.4*0.6 = 0.96$

$\text{sqrt}(1-0.96) = \text{sqrt}(0.04) = 0.2$

$r1 = (1 + 0.2)/0.8 = 1.2/0.8 = 1.5$ Hmm, let's try the factored approach:

$p*r^2 - r + q = 0$ can be factored as $(r - 1)(p*r - q) = 0$

[Check: $p*r^2 - p*r - q*r + q = p*r(r-1) - q(r-1) = (r-1)(p*r-q)$]

Roots: $r1 = 1$, $r2 = q/p = 0.6/0.4 = 3/2 = 1.5$

General solution (since r1 != r2):

$R(k) = A * 1^k + B * (3/2)^k = A + B*(3/2)^k$

c) Apply boundary conditions and compute R(2):

From R(0) = 0:

$A + B*(3/2)^0 = 0 \Rightarrow A + B = 0 \Rightarrow A = -B$

From R(5) = 1:

$A + B*(3/2)^5 = 1$

$-B + B*(3/2)^5 = 1$

$B*((3/2)^5 - 1) = 1$

$(3/2)^5 = 243/32 = 7.594$

B*(7.594 - 1) = 1 => B = 1/6.594 ~ 0.1517
A = -B ~ -0.1517

Compute R(2):
R(2) = A + B*(3/2)^2 = -B + B*(9/4) = B*(9/4 - 1) = B*(5/4)
R(2) = (5/4) / ((3/2)^5 - 1) = (5/4) / (243/32 - 1)
= (5/4) / (211/32) = (5/4)*(32/211) = 160/844 = 40/211 ~ 0.190

Fair-game value: 2/5 = 0.400
R(2) ~ 0.190 < 0.400: the biased game gives the gambler far less than a fair chance. With unfavorable odds (p < q), the gambler is much more likely to be ruined before winning.

Q2 Solution

a) Particular solution F_p(k) = c*k:

The full recurrence is:
p*F(k+1) - F(k) + q*F(k-1) = -1

Substitute F_p(k) = c*k (try linear since the homogeneous part already has r=1 as a root; but let's check if c*k works as a particular soln):

p*c*(k+1) - c*k + q*c*(k-1) = -1
c*k*(p - 1 + q) + c*(p - q) = -1
c*k*(p + q - 1) + c*(p - q) = -1
Since p + q = 1: c*k*0 + c*(p - q) = -1
c*(p - q) = -1
c = -1/(p - q) = -1/(0.4 - 0.6) = -1/(-0.2) = 5

Particular solution: F_p(k) = 5k

b) General solution and boundary conditions:

General solution (homogeneous + particular):
F(k) = A + B*(3/2)^k + 5k

Apply F(0) = 0:

A + B*(3/2)^0 + 5*0 = 0

A + B = 0 => A = -B

Apply F(5) = 0:

A + B*(3/2)^5 + 5*5 = 0

-B + B*(243/32) + 25 = 0

B*(243/32 - 1) = -25

B*(211/32) = -25

B = -25 * 32/211 = -800/211

A = 800/211

c) Compute F(2) and compare:

F(2) = A + B*(3/2)^2 + 5*2

= 800/211 + (-800/211)*(9/4) + 10

= 800/211 - 1800/211 + 10

= -1000/211 + 10

= -1000/211 + 2110/211

= 1110/211

~ 5.26 steps

Unbiased comparison: F_unbiased(2) = 2*(5-2) = 6 steps.

The biased game ends FASTER on average (5.26 vs 6 steps). This makes intuitive sense: with unfavorable odds the gambler is quickly ruined, so the game tends to end sooner (but almost always with the gambler losing).

Q3 Solution

a) Characteristic equation and roots:

Substitute F(n) = r^n into F(n) = F(n-1) + F(n-2):

r^n = r^(n-1) + r^(n-2)

Divide by r^(n-2): r^2 = r + 1

r^2 - r - 1 = 0 <-- characteristic equation

Quadratic formula:

r = (1 +/- sqrt(1 + 4)) / 2 = (1 +/- sqrt(5)) / 2

r1 = (1 + sqrt(5)) / 2 ~ (1 + 2.236) / 2 ~ 1.618 (golden ratio phi)

r2 = (1 - sqrt(5)) / 2 ~ (1 - 2.236) / 2 ~ -0.618

General solution:

F(n) = A * ((1+sqrt(5))/2)^n + B * ((1-sqrt(5))/2)^n

b) Apply F(0)=0 and F(1)=1 to find A and B (Binet's formula):

From F(0) = 0:

A * 1 + B * 1 = 0 => A + B = 0 => B = -A

From F(1) = 1:

A * (1+sqrt(5))/2 + B * (1-sqrt(5))/2 = 1

A * (1+sqrt(5))/2 - A * (1-sqrt(5))/2 = 1

A * [(1+sqrt(5)) - (1-sqrt(5))] / 2 = 1

A * [2*sqrt(5)] / 2 = 1

A * sqrt(5) = 1

A = 1/sqrt(5), B = -1/sqrt(5)

Binet's formula:

F(n) = (1/sqrt(5)) * [((1+sqrt(5))/2)^n - ((1-sqrt(5))/2)^n]

c) Convergence of F(n+1)/F(n) to the golden ratio phi:

Using Binet's formula:

F(n+1)/F(n) = [phi^(n+1) - psi^(n+1)] / [phi^n - psi^n]

where phi = (1+sqrt(5))/2 ~ 1.618 and psi = (1-sqrt(5))/2 ~ -0.618

Divide numerator and denominator by phi^n:

F(n+1)/F(n) = [phi - psi*(psi/phi)^n * phi] / [1 - (psi/phi)^n]

More cleanly, divide top and bottom by phi^n:

= (phi - (psi/phi)^n * psi) / (1 - (psi/phi)^n)

Now |psi/phi| = |(-0.618)/(1.618)| ~ 0.382 < 1.
Therefore (psi/phi)^n -> 0 as n -> infinity.

In the limit:
F(n+1)/F(n) -> phi / 1 = phi = (1+sqrt(5))/2 (OK)

The psi term decays to zero because |psi| = 0.618 < 1, meaning
psi^n -> 0 exponentially. For large n, F(n) ~ phi^n / sqrt(5),
and so the ratio F(n+1)/F(n) ~ phi^(n+1)/phi^n = phi.